

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0353

Herdlyn 138 kV
February 25, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0353. The Interconnection Customer proposes a 112.0 MW natural gas combined cycle facility interconnecting at the Herdlyn 138 kV substation in the Maryland area. The request was received on July 02, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 186 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,046,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of ROSSMOOR SOUTH – TRAMWAY 138kV Line, the SLIGO – DAIEN 138kV loads to 121.8% of its 261 MVA rating. Project DFAX: 5.9%.

For the loss of TRAMWAY – ROSSMOOR S 138kV line, the LINDEN – SLIGO 138kV loads to 118.1% of its 293 MVA rating. Project DFAX: 5.9%.

The BNKER HILL – BLTSVILLE 138kV line is overloaded to 116.3% of its 586 MVA emergency rating for the loss of the TABER – GLENANNN 345kV. The project contributes 7.7% DFAX (8.6 MW) to this overload.

The DARIEN – COLESVILLE 138kV is overloaded to 115.8% of its 366 MVA emergency rating for the loss of the TABER – GLENANNAN 345kV. The project contributes 5.9% DFAX (6.7 MW) to this overload.

The BUNKER HILL – NASA 138kV is overloaded to 114.3% of its 629 MVA emergency rating for the loss of the GLENANNAN – TABER 345kV line. The project contributes 7.7% DFAX (8.6 MW) to this overload.

The TABER – GLENANNAN 345kV is overloaded to 111.3% of its 1,656 MVA emergency rating for the loss of the SIERRA PACIFIC LUMBER 138/1 kV TX. The project contributes 22.4% DFAX (25.0 MW) to this overload.

The JEROME – HIGH RIDGE 138kV Line is overloaded to 111.1% of its 601 MVA emergency rating for the loss of the TABER – DIXLETA 345kV L. The project contributes 10.2% DFAX (11.4 MW) to this overload.

For the loss of TABER – DIXLETA 345kV, the DIXLETA – N POLE 138kV L loads to 109.1% of its 338 MVA rating. Project DFAX: 6.2%.

The TABER – DIXLETA 345kV is overloaded to 108.3% of its 1,619 MVA emergency rating for the loss of the SIERRA PACIFIC LUMBER 138/1 kV Auto. The project contributes 20.4% DFAX (22.9 MW) to this overload.

For the loss of TOMBSTONE – BIXBYHILL 500kV, the SHARYLAND – BIXBY HILL 345kV loads to 107.9% of its 1,851 MVA rating. Project DFAX: 7.8%.

The TOMBSTONE – SHARYLAND 345kV Line is overloaded to 107.1% of its 1,905 MVA emergency rating for the loss of the TOMBSTONE – BIXBY HILL 500kV line. The project contributes 7.6% DFAX (8.5 MW) to this overload.

The LAKE CREEK – SIERRA PACIFIC LUMBER 138kV is overloaded to 106.7% of its 612 MVA emergency rating for the loss of the TABER – GLENANNAN 345kV. The project contributes 5.7% DFAX (6.4 MW) to this overload.

PHILLIPS PUMP 1 – BEVERLY FARMS 138kV line: 106.1% loading (205 MVA) following the outage of the GLENANNAN – TABER 345kV; DFAX = 5.2%.

SIERRA PACIFIC LUMBER 345/1 kV XFMR: 103.9% loading (963 MVA) following the outage of the TABER – GLENANNAN 345kV; DFAX = 7.5%.

SIERRA PACIFIC LUMBER 138-1 kV XFMR: 103.9% loading (963 MVA) following the outage of the TABER – GLENANNAN 345kV; DFAX = 7.5%.

For the loss of SIUC – FAIRLAND 138kV, the JEROME – FAIRLAND 138kV loads to 103.7% of its 233 MVA rating. Project DFAX: 5.3%.

FAIRLAND – SUNBRIGHT 138kV: 102.6% loading (673 MVA) following the outage of the GLENANNAN – TABER 345kV; DFAX = 11.7%.

The COLESVILLE – SUNBRIGHT 138kV line is overloaded to 101.3% of its 736 MVA emergency rating for the loss of the TABER – GLENANNAN SUBSTATION 345kV. The project contributes 11.7% DFAX (13.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SLIGO – DARIEN 138kV Line	Rebuild line with higher-capacity conductor	\$4,114,000	10% (\$428,000)
SLIGO – LINDEN 138kV Line	Rebuild line with higher-capacity conductor	\$811,000	10% (\$84,000)
BELTSVILLE – BUNKER HILL 138kV	Rebuild line with higher-capacity conductor	\$922,000	1% (\$12,000)
COLESVILLE – DARIEN 138kV	Rebuild line with higher-capacity conductor	\$843,000	3% (\$28,000)
NASA – BUNKER HILL 138kV	Rebuild line with higher-capacity conductor	\$3,169,000	1% (\$41,000)
TABER – GLENANNAN 345kV	Rebuild line with higher-capacity conductor	\$37,265,000	1% (\$336,000)
HIGH RIDGE SUBSTATION – JEROME 138kV	Rebuild line with higher-capacity conductor	\$2,675,000	1% (\$28,000)
NORTH POLE – DIXLETA 138kV #1	Reconductor line	\$282,000	1% (\$4,000)
TABER – DIXLETA 345kV	Reconductor line	\$6,867,000	1% (\$61,000)
SHARYLAND – BIXBY HILL 345kV Line	Reconductor line	\$17,178,000	1% (\$127,000)
TOMBSTONE – SHARYLAND 345kV	Reconductor line	\$21,986,000	1% (\$160,000)
LAKE CRK – SIERRA PACIFIC LUMBER 138kV line	Reconductor line	\$926,000	1% (\$13,000)
PHILLIPS PUMP 1 – BEVERLY FARMS 138kV line	Reconductor line	\$794,000	13% (\$107,000)
SIERRA PACIFIC LUMBER 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$15,507,000	1% (\$166,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SIERRA PACIFIC LUMBER 138 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$4,317,000	1% (\$46,000)
FAIRLAND SUBSTATION – JEROME 138kV	Reconductor line	\$2,582,000	15% (\$398,000)
FAIRLAND – SUNBRIGHT 138kV	Reconductor line	\$115,000	1% (\$2,000)
COLESVILLE – SUNBRIGHT 138kV line	Reconductor line	\$391,000	1% (\$5,000)
	Total allocated to project		\$2,046,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.